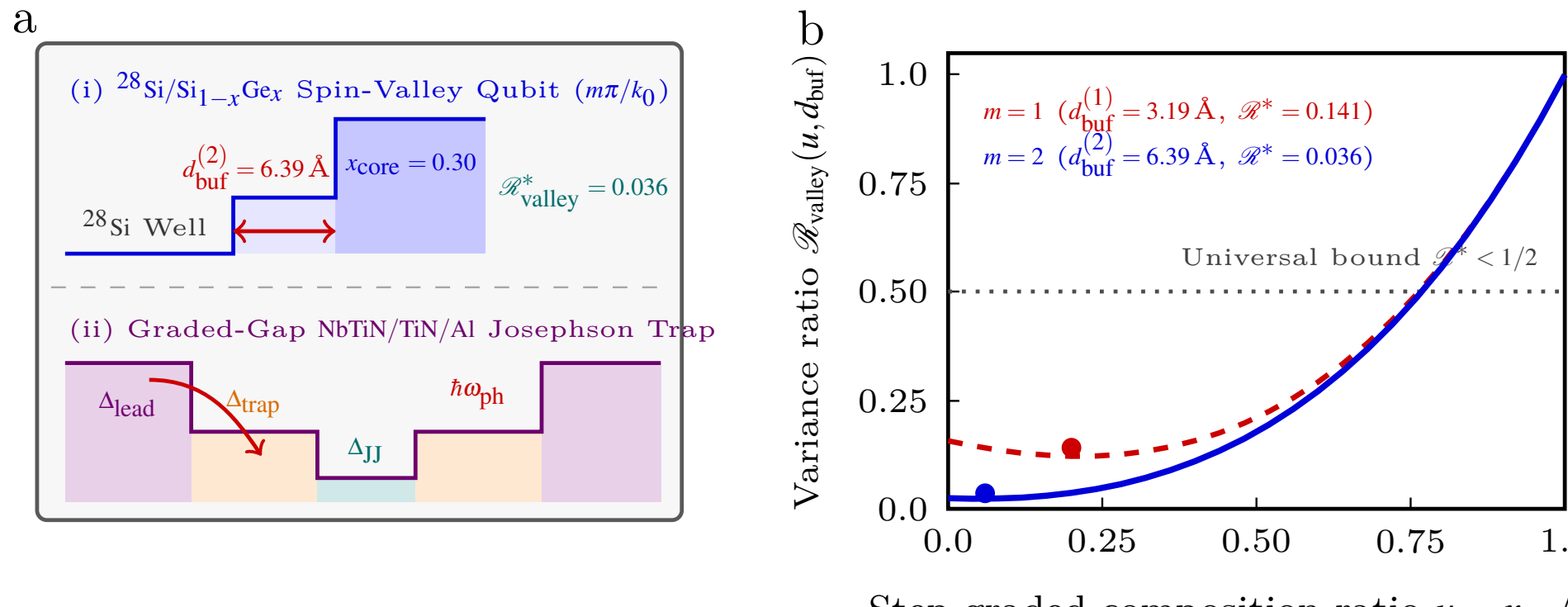
a
(i) ^{28}Si/Si$_{1-x}$Ge$_x$ Spin-Valley Qubit ($m\pi/k_0$)
$d_{\mathrm{buf}}^{(2)} = 6.39$ Å
$x_{\mathrm{core}} = 0.30$
$\mathscr{R}^*_{\mathrm{valley}} = 0.036$
^{28}Si Well
(ii) Graded-Gap NbTiN/TiN/Al Josephson Trap
Δ_{lead}
Δ_{trap}
Δ_{JJ}
$\hbar\omega_{\mathrm{ph}}$
b
Variance ratio $\mathscr{R}_{\mathrm{valley}}(u, d_{\mathrm{buf}})$
$m = 1$ ($d_{\mathrm{buf}}^{(1)} = 3.19$ Å, $\mathscr{R}^* = 0.141$)
$m = 2$ ($d_{\mathrm{buf}}^{(2)} = 6.39$ Å, $\mathscr{R}^* = 0.036$)
Universal bound $\mathscr{R}^* < 1/2$
1.0
0.75
0.50
0.25
0.0
0.0
0.25
0.50
0.75
1.